

Unit Economics & Pricing

Cost per query, margin analysis, and recommended pricing tiers — based on actual LLM API costs.

Cost per query

Query Type	LLM Cost	Mix	Weighted Cost
Single query	\$1.00	20%	\$0.20
Multi query (full pipeline)	\$4.00	80%	\$3.20
Blended cost per question			\$3.40

Per-user cost by usage level

User Type	Questions / Month	Your Cost / User
Light	15	\$51
Moderate	40	\$136
Heavy	80	\$272

Recommended pricing

Usage caps are critical. Without them, a heavy user at 150 questions/month costs you \$510 and destroys margin on a \$499 plan. Overage pricing provides a safety net and scales revenue with usage.

Plan	Price	Included Questions	Your Cost (Maxed)	Margin
Starter	\$499/mo	50	\$170	66%
Growth	\$1,499/mo	200	\$680	55%
Enterprise	Custom	Custom	Volume-based	50%+
Overage	\$12/question	—	\$3.40	72%

Concurrency

Users never queue behind each other. Every request flows through independent, parallel paths.

Node.js API

Async, non-blocking I/O. A single process handles hundreds of concurrent in-flight requests without waiting.

LLM API (OpenRouter)

Stateless HTTP requests. Charge per token, not per connection. 10 simultaneous queries cost the same as 10 sequential ones.

PostgreSQL

Connection pooling via PgBouncer. Each user's SQL runs independently. Handles 50+ concurrent queries easily.

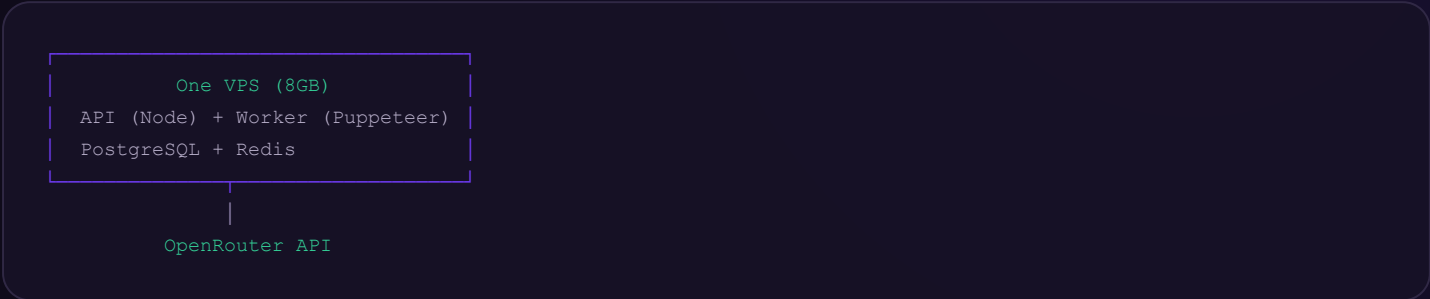
PDF rendering

Puppeteer is memory-heavy. At scale, move to a dedicated worker with BullMQ queue to avoid starving the API server.

10 → 100 → 1,000 users

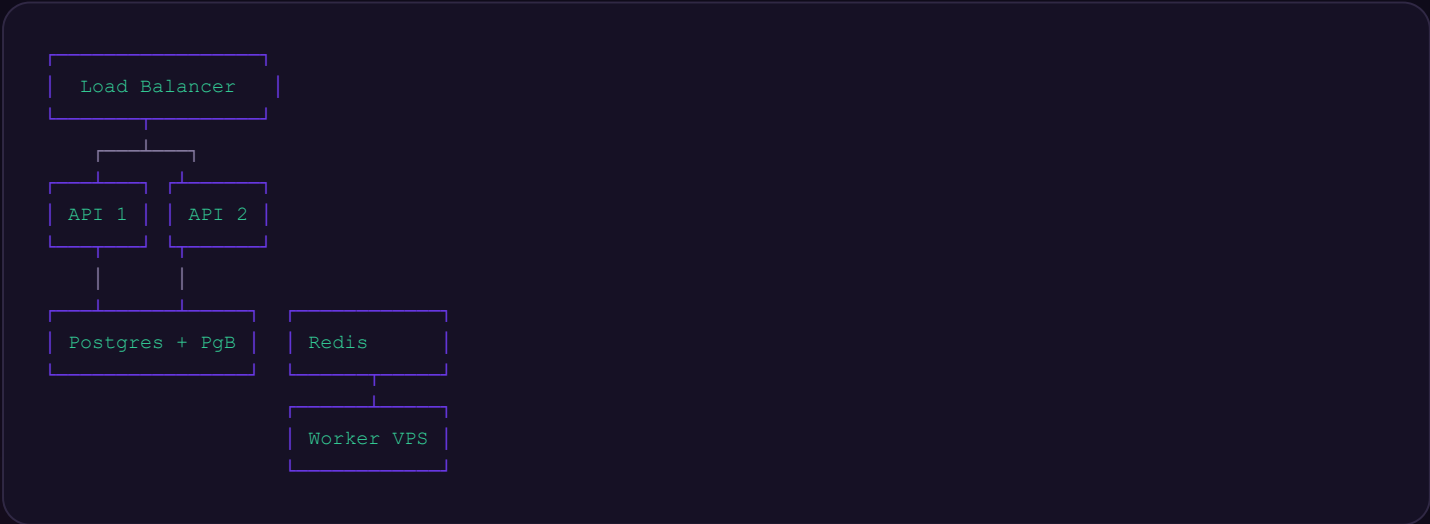
Same stack (Node.js, PostgreSQL, Redis, OpenRouter, Puppeteer). Scale by spending more, not rewriting.

10 users — Single box



ITEM	SPEC	COST/MONTH
VPS	4 vCPU, 8GB RAM	\$40
Managed Postgres	Shared, 1GB	\$15
OpenRouter LLM	~500 queries	~\$1,700
Total		~\$1,760
Revenue (10 x \$499)		\$4,990
Margin		65%

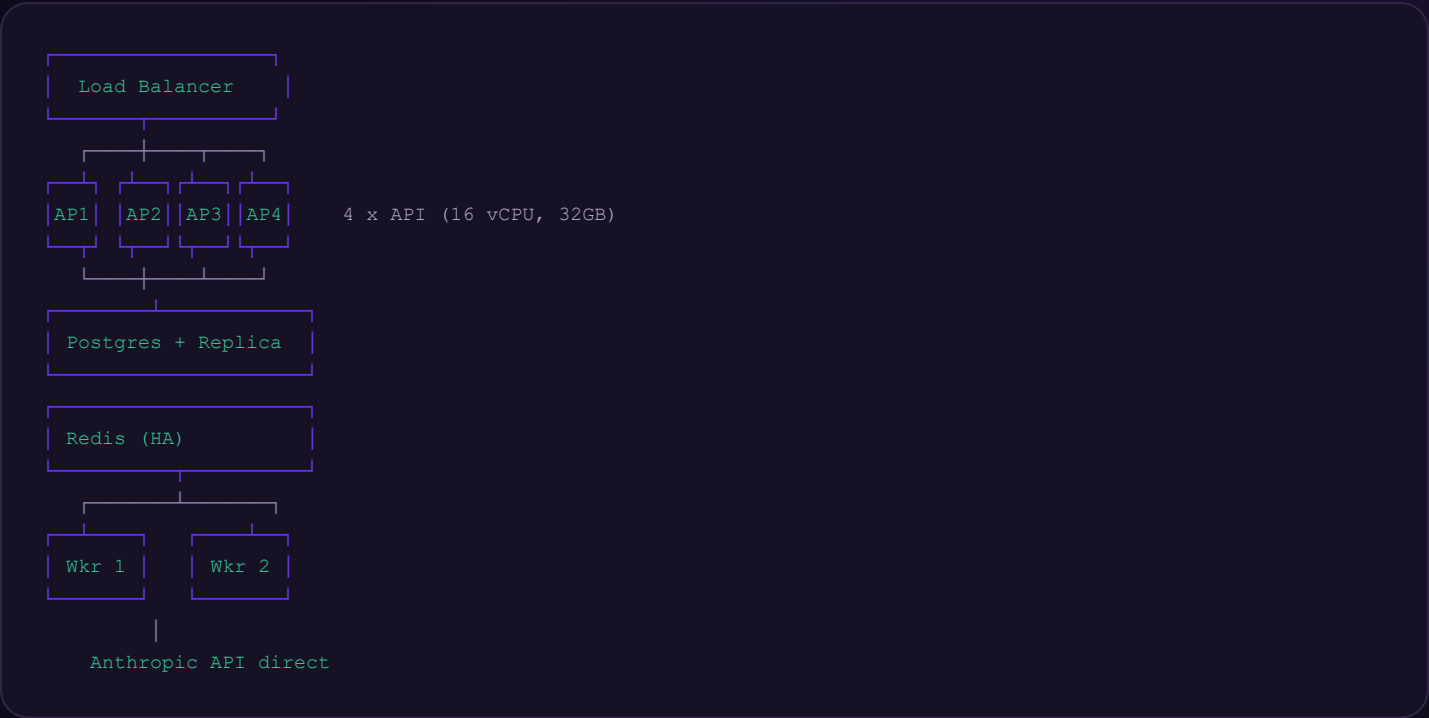
100 users — Two servers + dedicated DB



ITEM	SPEC	COST/MONTH
Load balancer	nginx / ALB	\$20
2 x API servers	8 vCPU, 16GB each	\$160
Managed Postgres	Dedicated, 4GB	\$80
Redis	1GB managed	\$20
Worker VPS	4 vCPU, 16GB	\$80
OpenRouter LLM	~5,000 queries	~\$17,000

ITEM	SPEC	COST/MONTH
Total		~\$17,370
Revenue (100 x \$700 avg)		\$70,000
Margin		75%

1,000 users — Four API servers + read replica



ITEM	SPEC	COST/MONTH
Load balancer	Managed	\$25
4 x API servers	16 vCPU, 32GB each	\$640
Postgres (primary)	8 vCPU, 32GB	\$400
Read replica	4 vCPU, 16GB	\$200
Redis (HA)	4GB managed	\$50
2 x Worker VPS	8 vCPU, 32GB each	\$320
Monitoring	Grafana / Datadog	\$100
LLM API	~50,000 queries	~\$170,000
Total		~\$171,735
Revenue (1,000 x \$800 avg)		\$800,000
Margin		78%

LLM rate limits — The real scaling bottleneck

Infrastructure costs are 1-2% of total spend. LLM API is 98%+. Rate limits, not server capacity, are what you hit first.

RATE LIMIT TIER	REQUESTS / MIN	SUPPORTS
Default (free)	60 RPM	~10 users
Standard	200 RPM	~50 users
Pro / custom	500+ RPM	100+ users
Anthropic direct	Negotiated	1,000+ users

Scaling rule

API slow?

Upgrade box to 32GB or add another instance.

DB slow?

Upgrade Postgres tier. Add read replica.

PDFs backing up?

Upgrade worker or add a second one.

Rate limited?

Request increase or switch to Anthropic direct.

Finding customers

All channels excluding LinkedIn outreach. Prioritized by cost, speed, and expected ROI.

Tier 1 — Free, start immediately

Cold email (targeted, not spam)

- Buy 500 CFO / Head of Analytics contacts (Apollo.io, \$50/mo)
- 3-email sequences via Instantly.ai (\$30/mo)
- Lead with: "How long does it take your team to answer a board question?"
- Target: 2-3% reply rate = 10-15 conversations

Communities

- Reddit: r/analytics, r/BusinessIntelligence, r/datascience
- Slack: dbt, Locally Optimistic, Data Engineering, RevOps Co-op
- Indie Hackers: build story, revenue milestones
- Answer real questions. People check your profile.

Hacker News — Show HN

- Title: "Show HN: Ask your database anything, get a consultant-grade report"
- Post Tuesday/Wednesday, 8-9am ET
- Engage with every comment
- HN audience = technical founders who buy tools

Twitter / X — Build in public

- Screenshots, architecture, customer wins
- Engage data/analytics Twitter
- Contrarian takes on analytics hiring
- Consistent posting compounds

Tier 2 — Small spend (\$500-1,500/mo), weeks 2-4

Google Ads (high-intent search)

- Keywords: "automated business reporting", "AI data analyst", "alternative to hiring data analyst"
- Budget: \$500-1,000/month
- Dedicated landing page with one CTA: book a pilot
- Expected: \$20-60/click, 5-10% conversion = \$200-600/lead

Content / SEO

- 5-10 articles targeting buyer keywords
- "How to automate executive reporting"
- "AI vs hiring a data analyst"
- Host on blog.claritect.ai — compounds over time

YouTube demos

- 3-5 minute screen recordings of real questions answered
- Cross-post as Twitter threads with screenshots
- A 3-min demo converts better than 10 pages of copy

Newsletter sponsorships

- Data Science Roundup (~\$200-500/issue)
- Analytics Engineering Roundup
- Lenny's Newsletter (\$5K+, premium audience)
- One good sponsorship = 20-50 qualified visits

Tier 3 — Bigger spend, month 2+

AppSumo / marketplaces

Lifetime deal at \$79-149 to get first 50-100 users. Lose per user but gain: reviews, testimonials, social proof for full-price sales.

Partnerships

Partner with dbt consultancies, data agencies, fractional CFO firms. 20% revenue share. 10 partners x 1 client/month = 10 customers.

Live webinars

Monthly "Ask your data anything — live" event. Connect sample DB, take audience questions. Record and reuse as content.

Product Hunt launch plan

Product is ready. Here's the execution plan for a top-10 launch day.

2 weeks before — Build support network & prepare assets

Network

- Create maker profile, fill it out completely
- Get 5-10 friends to create PH accounts now
- Post in Indie Hackers, Twitter: "We're launching on [date]"
- Find a hunter with a large PH following
- Prepare email to everyone you know

Assets

- Tagline (60 chars): "Ask your data anything. Get consultant-grade answers."
- 5-6 gallery images (1270x760px): hero shot, before/after, pipeline, comparison, sample report, pricing
- Video (1-2 min): screen recording, real question, real report, captions
- First comment: origin story, honest, human
- Special offer: "PH community: first month free" or "\$299/mo lifetime for first 50"

1 week before

- Have 3 people who've never seen Claritect try the full flow (homepage → pilot → demo)
- Fix any friction they hit
- Write launch-day tweets (3-4 throughout the day)
- Draft launch-day email to your network
- Pick launch day: Tuesday or Wednesday (avoid Monday crowding, Friday low traffic)

Launch day schedule

TIME (PT)	ACTION
12:01 AM	Product goes live on Product Hunt
12:05 AM	Post your maker comment (origin story)
6:00 AM	Send email to your network with link
7:00 AM	Tweet thread: "Today we launched Claritect on PH..."
8:00 AM	Post in communities (Indie Hackers, Reddit, Slack groups)
9:00 AM+	Reply to every PH comment within 15 minutes
12:00 PM	Second tweet: share early traction / comments
6:00 PM	Thank-you tweet with screenshots

After launch

DAY	ACTION
Day 2	Write "what we learned from our PH launch" for Indie Hackers / Twitter
Day 3	DM everyone who commented — offer a free pilot
Week 1	Add PH badge to website
Week 2	Turn PH comments into testimonials. Submit Show HN.
Month 1	Blog post: "How we got X upvotes on Product Hunt"

Realistic expectations

